

Kaiming Liu

Updated: October 2, 2026

Email: kl54@illinois.edu

GitHub: github.com/kl543

Homepage: kl543.github.io

Research Profile	Probabilistic machine learning for multimodal conditional inference and robustness under distribution shift, tested on scientific inverse problems.	
Education	University of Illinois Urbana-Champaign <i>Ph.D. in Physics, in progress</i> Advisor: Bryan K. Clark	Aug 2023–present
	Xi'an Jiaotong University <i>B.S. in Physics (Honors)</i> <i>Special Class for the Gifted Young</i>	Sep 2019–Jul 2023 Sep 2017–Jul 2019
	University of California, Berkeley <i>BPIE Program (non-degree visiting program)</i>	Aug–Dec 2021
Current Research	University of Illinois Urbana-Champaign <i>Graduate researcher, Bryan K. Clark group</i>	Nov 2025–present
Multimodal inference / crystal-field parameters		
• Built residual-MLP conditional Gaussian mixtures to represent multiple plausible crystal-field solutions from susceptibility. • Validated candidates through forward reconstruction; studied noise, scale, and background shifts and experimental measurements. • Evaluated conditional Gaussian mixtures and conditional RealNVP normalizing flows in a pilot density-model comparison.		
Robust inverse prediction / neutron scattering		
• Trained MLP and ResNet conditional Gaussian mixtures for high-dimensional inverse prediction of exchange parameters from neutron maps. • Used exact Sunny forward evaluation and classical fit references; tested noise, intensity-scale, and background shifts.		
Neural-network backflow / JAX and Monte Carlo		
• Implemented neural-network backflow and variational Monte Carlo in JAX for Hubbard models, with GPU/multi-GPU sampling. • Profiled local-energy evaluation and gradient synchronization; verified chunked evaluation and global reductions against reference computations.		

Publications	[P1] Evidence of intertwined pair density and charge density wave orders in UTe₂ Zhen Zhu, Yudi Huang, Julian May-Mann, Kaiming Liu, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan. <i>PNAS</i> 123 (29), e2602117123 (2026). arXiv.	
Prior Research	University of Illinois Urbana-Champaign <i>Research assistant, Vidya Madhavan group</i> Aug 2023–Oct 2025 • Grew thin-film heterostructures with MBE and characterized electronic structure with STM/STS in ultrahigh vacuum. • Automated growth temperature control and built Python/MATLAB tools for quasiparticle-interference analysis. Xi'an Jiaotong University <i>Research assistant, Yongchang Zhang group</i> Mar 2022–Jul 2023 • Simulated Rydberg-atom dynamics and photon-mediated interactions in Python/MATLAB to study quantum nonlinear optics. • Built a discrete-time lattice model of two-qubit phase-flip dynamics and many-body regimes. University of California, Berkeley <i>Research assistant, Michael F. Crommie group</i> Oct–Dec 2021 • Assisted synthesis and STM characterization of graphene nanoribbons with five-membered rings to study their electronic states.	
Teaching	University of Illinois Urbana-Champaign <i>Teaching assistant, PHYS 212</i> Spring & Fall 2026 <i>Teaching assistant, PHYS 102</i> Aug–Dec 2023	
Selected Honors	China National Scholarship 2021 Xi'an Jiaotong University Mathematical Contest in Modeling, Meritorious Winner 2021 COMAP	
Technical Skills	<i>ML / modeling</i>: PyTorch, JAX; conditional mixtures, residual networks, probabilistic inference. <i>Research engineering</i>: Python, NumPy, SciPy, Git, Linux; GPU/multi-GPU and cluster computation, runtime profiling. <i>Numerical / scientific methods</i>: MCMC, VMC, numerical optimization; forward-model evaluation and simulation-to-experiment robustness. <i>Experimental physics</i>: MBE, STM/STS, UHV; feedback control, automation, MATLAB and data analysis.	